

TEZPUR UNIVERSITY
Autumn Semester End Examination, 2023
CS621: MOBILE COMPUTING

Full Marks:45

Time:2 Hrs 40 mins

Figures in the right hand indicate marks for the individual question.

Part B

1. What are the purposes of having 4 addresses in IEEE 802.11 MAC frame, instead of having only two addresses like in IEEE 802.3? Briefly explain how power saving mode for IEEE 802.11 (infrastructure based) works. [2+2=4]
2. How is the concept of master-slave based communication used in Piconet? What is Scatternet? In Bluetooth, what is the sequence of control packets exchanged and how the potential Master and Slave nodes change states to form a Piconet? [1+1+3=5]
3. Suppose a Piconet is formed with a Master and a Slave. Compute the maximum ACL data rate in each direction (that is, master to slave and slave to master) that can be supported to a Slave, assuming single DH3 connection per Slave. [Hints: assume 253 bits for circuit settling, 72 bits for Access Code and 54 bits for Packet Header, 16 bits for Payload Header and 16 bits for CRC] [5]
4. Suppose 3 stations ST1, ST2 and ST3 in a WLAN have frame arrived at some instance of time for transmission over the wireless media using CSMA/CA MAC protocol as shown below.
 - For ST1, packet arrives at time 0, with size=1000bytes
 - For ST2, packet arrives at time 100 μ s, with size=1400bytes
 - For ST3, packet arrives at time 200 μ s, with size=500bytes

Given: slot time=20 μ s, SIFS=10 μ s, DIFS=50 μ s, RTS=CTS=ACK =100bytes, RTS Threshold=1200bytes, Fragmentation Threshold = 2000bytes, backoff=2 and the WLAN is operating at data rate of 200bytes per slot time. When will the data transfer by all the stations be completed? [6]
5. In mobile-IP –
 - a. How does a MN perform registration after visiting a foreign network? [2]
 - b. How does a mobile node (MN) visiting a foreign network (FN) receive packet from a Correspondent Node (CN) destined for mobile node's home IP? [2]
 - c. Why is reverse tunneling required in mobile IP and how does reverse tunneling work? [2]
6. Why is conventional TCP not suitable for wireless network? Briefly explain how I-TCP works in wireless network considering packets flow from FH to MN and MN to FH. Give a merit and a demerit of using I-TCP. [1+4+2=7]
7. How is routing in MANET different from routing in Internet? What do you mean by Proactive and Reactive Routing protocols? Mention their relative merits and demerits. Briefly discuss one method to reduce route discovery flooding in Routing. [1+2+2+1=6]
8. Briefly explain the Route Discovery Process in Dynamic Source Routing (DSR) protocol using RREQ and RREP packets. What are the optimizations possible through route caching? Mention two main drawbacks of DSR. [3+2+1=6]